Minglai Yang

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Working Experience

Scale AI **San Francisco, CA**
Research Scientist, Agents *Jun 2026 – Present*
- Working on agents and RL environments.
- Led Rubric Dropout [6], a simple method to mitigate reward hacking in rubric-as-reward RL (**Scale Labs**).
- Led From Copying to Composition [1], a controlled study of how language models develop in-context learning across 17 models (**Scale Labs**).
- Co-authored research on evidence-first agentic verification [3], research environments for recursive self-improvement [4], and reliable enterprise agent deployment [5] (**Scale Labs**).

Abaka AI **Palo Alto, CA**
Senior Member of Technical Staff - Agents Lead *Jan 2026 – Jun 2026*
- Led a 15-person cross-functional team on an MCP-based agent RL project, coordinating environment development, synthetic/real data generation, expert annotation, and agent evaluation workstreams.
- Designed environment simulators spanning B2B and B2C RL environments and deployed Harbor for systematic agent benchmarking and evaluation.
- Built a synthetic task and rubrics generation pipeline, producing over 3,000 high-quality, long-horizon tasks with automated evaluation criteria across 10 environments with ∼800 tools in total.
- Led Dr. DocBench [8], overseeing evaluation design, data selection, and human annotation pipeline (**EMNLP In Submission**).
- Co-authored research on instruction-guided video editing evaluation [9] (**COLM In Submission**) and LLM error verifiability [10] (**COLM In Submission**).

Coretechs **Kensington, MD**
Machine Learning Engineer Intern (Full Time) *May 2024 – Aug 2024*
- Developed GPT-powered Slack bots with retrieval-augmented generation (RAG) and integrated real-time Slack chat into the official website using JavaScript, increasing active user engagement by 57%.
- Led end-to-end website development, from design to deployment, leveraging AWS (EC2, RDS, S3), Nginx, and Docker for scalable infrastructure.
- Conducted performance testing with Selenium, simulating 500+ users to ensure system reliability under high load.
- Received the *Best Intern Award* for outstanding contributions to the company's first GenAI project.

The University of Arizona, AI Core **Tucson, AZ**
Lead AI Engineer (Part Time) *Feb 2024 – Dec 2024*
- Built tool-calling AI agents with Retrieval-Augmented Generation (RAG) for internal knowledge; trained a custom long-context documentation QA model, improving top-1 accuracy by 48% on internal benchmarks.
- Designed and shipped an end-to-end campus chatbot website serving 50,000+ UA students with long-context retrieval; increased student web engagement by $5.39\times$ post-launch.

Research Experience

University of Arizona, CLU-LAB **Tucson, AZ**
Undergraduate Research Assistant *Oct 2024 – Dec 2025*

- Advisor: Prof. Mihai Surdeanu, Dr. Liangming Pan and Dr. Steven Bethard
- Led Research Project [1] on analyzing LLM reasoning, designed GSM-DC, a controlled benchmark for evaluating language models reasoning under irrelevant context, revealing **6 new empirical laws** on how irrelevant context disrupts reasoning, which training regimes improve robustness, and proposing test-time methods that achieve up to **6.3%** out-of-distribution performance gains (**EMNLP Oral – First Author**).
- Conducting research with Dr. Vikas Yadav on speculative decoding strategies for LLMs, introducing CopySpec [2] , a copy-and-verify approach that delivers up to **3.08**× **faster generation** and **2.66**× **higher efficiency** in LLM self-correction reasoning (**EMNLP Oral - Second Author**).
- Leading a project with Prof. Steven Bethard on long-context LLM reasoning, investigating whether model reasoning is trustworthy versus prone to hallucination and other failure modes (**SemEval In Submission - First Author**).
- Collaborated with Dr. Chicheng Zhang to develop LLM-guided warm starts for RL [3], achieving at least $20\times$ higher data efficiency (**EXAIT@ICML Workshop - Second Author**).
- Proposed using PPO to optimize the HIT@DICT reward for relation extraction; our RL optimization raised one-shot NYT29 F1 **from 24.65% to 48.11%**, with more stable convergence (**ICLR In Submission - Third Author**).

University of Arizona, ML4AI-LAB and IVI-LAB **Tucson, AZ**
Undergraduate Research Assistant *Oct 2023 – Dec 2025*

- Advisor: Prof. Kobus Barnard and Dr. Adarsh Pyarelal
- Developing dynamic Bayesian networks and Theory of Mind models within the ToMCAT project to improve AI modeling of human coordination.
- Leading a project on evaluating LLM planning capabilities under adversarial conditions, designing a controlled testbed to conduct stepwise analysis and benchmarking.
- Designing probabilistic models to disentangle causal interpersonal coordination from correlation.
- Co-authored a $544K, two-year research proposal, focusing on LLM planning, temporal-causal reasoning, and robustness evaluation against noisy commands.

Education

University of Arizona **Tucson**
B.S. in Computer Science *2023–2025*

- GPA: *4.0 / 4.0*
- Graduated December 2025, **summa cum laude** (accelerated completion of 4-year curriculum in 2 years)
- All relevant coursework available here.

Publications

1. *How Is LLM Reasoning Distracted by Irrelevant Context? An Analysis Using a Controlled Benchmark*
   Minglai Yang, Ethan Huang, Liang Zhang, Mihai Surdeanu, William Wang, Liangming Pan
   EMNLP 2025 — Main Conference **[Oral]** (CORE A*), 2025 [preprint] [code] [slides] [poster] [project page]
2. *CopySpec: Accelerating LLMs with Speculative Copy-and-Paste Without Compromising Quality*
   Razvan Dumitru, **Minglai Yang**, Vikas Yadav, Mihai Surdeanu
   EMNLP 2025 — Main Conference **[Oral]** (CORE A*), 2025 [preprint] [code]
3. *Improving the Data-efficiency of Reinforcement Learning by Warm-starting with LLM*
   Thang Duong, **Minglai Yang**, Chicheng Zhang
   Exploration in AI Today (EXAIT) Workshop @ ICML, 2025 [preprint] [code]
4. *AlignSAE: Concept-Aligned Sparse Autoencoders*
   Minglai Yang, Xinyu Guo, Zhengliang Shi, Jinhe Bi, Steven Bethard, Mihai Surdeanu, Liangming Pan
   Transactions on Machine Learning Research (TMLR), 2026 [openreview] [preprint] [code] [project page]
5. *EchoRL: Reinforcement Learning via Rollout Echoing*
   Jinhe Bi, Aniri, **Minglai Yang**, Xingcheng Zhou, Wenke Huang, Sikuan Yan, Yujun Wang, Zixuan Cao, Michael Färber, Xun Xiao, Volker Tresp, Yunpu Ma
   International Conference on Machine Learning (ICML), 2026 [preprint]

Preprints

1. *From Copying to Composition: How Language Models Develop In-Context Learning*
   Minglai Yang, Xinyu Guo, Renxiong Wang, Sunjie Hou, Daniel Yue Zhang, Yunzhong He
   Scale Labs, 2026 [preprint]

2. *CyberArena: Benchmarking AI Agents in Live Symmetric Attack-and-Defense Combat*
   Bangya Liu, Jianhong Tu, Yueyang Pan, Hanhong Zhao, Zhe Ye, et al. (incl. **Minglai Yang**)
   arXiv preprint, 2026 [preprint]
3. *Evidence-First Agentic Verification*
   Reza Sanayei, Razvan-Gabriel Dumitru, Sunjie Hou, MohammadHossein Rezaei, Haris Riaz, Robert Vacareanu, **Minglai Yang**, Ying Liu, Mihai Surdeanu, Daniel Yue Zhang
   Scale Labs, 2026 [preprint]
4. *RSI-Forge: From Research Papers to Environments for Recursive Self-Improvement*
   Renxiong Wang, Darvin Yi, Abril Herrlein, Anas Mahmoud, Advait Gosai, et al. (incl. **Minglai Yang**)
   Scale Labs, 2026 [preprint]
5. *READY or Not: Reliable Enterprise Agent Deployment*
   Veronica Chatrath, Bryan Zhu, Jingxuan Fan, George Pu, Soham Dinesh Tiwari, et al. (incl. **Minglai Yang**)
   Scale Labs, 2026 [preprint]
6. *Rubric Dropout: A Simple Way to Mitigate Reward Hacking in Rubric-as-Reward RL*
   Minglai Yang, Xinyu Guo, Utkarsh Tyagi, Mian Zhang, Razvan Dumitru, Sunjie Hou, Yunzhong He, Daniel Yue Zhang, Ying Liu
   Scale Labs, 2026 [preprint]
7. *The Answer Lies Within: Self-Derived Rewards Enable Explainable Relation Extraction*
   Xinyu Guo, Zhengliang Shi, **Minglai Yang**, Mihai Surdeanu
   ICLR In Submission, 2026 [preprint][code]
8. *Dr. DocBench: A Comprehensive Benchmark for Expert-Level and Difficult Document Parsing*
   Minglai Yang*, Xinyan Velocity Yu*, Pengyuan Li, Xinyu Guo, Zhenting Qi, Konwoo Kim, Longtian Ye, Xiaolong Luo, Jinhe Bi, Henry Zhang, et al. (**equal contribution*)
   EMNLP In Submission, 2026 [preprint]
9. *VEFX-Bench: A Holistic Benchmark for Generic Video Editing and Visual Effects*
   Xiangbo Gao, Sicong Jiang, Bangya Liu, Xinghao Chen, **Minglai Yang**, Siyuan Yang, Mingyang Wu, Jiongze Yu, Qi Zheng, Haozhi Wang, Jiayi Zhang, Jie Yang, Zihan Wang, Qing Yin, Zhengzhong Tu
   COLM In Submission, 2026 [preprint]
10. *Justified or Just Convincing? Error Verifiability as a Dimension of LLM Quality*
    Xiaoyuan Zhu, Kimberly Le Truong, Riccardo Fogliato, Gokul Swamy, Weijian Zhang, **Minglai Yang**, Longtian Ye, Bangya Liu, Minghao Liu, Andrew Ilyas, Steven Wu
    COLM In Submission, 2026 [preprint]
11. *Opening the Black Box: A Survey on the Mechanisms of Multi-Step Reasoning in Large Language Models*
    Liangming Pan, Jason Liang, Jiaran Ye, **Minglai Yang**, Xinyuan Lu, Fengbin Zhu
    arXiv preprint, 2026 [preprint]
12. *Knowledge Index of Noah's Ark*
    Sheng Jin, Minghao Liu, Yunze Xiao, et al. (incl. **Minglai Yang**)
    arXiv preprint, 2026 [preprint]
13. *Triaging Threats to Specialized Guardrails*
    Wenjie Jacky Mo, Xiaofei Wen, Rui Cai, Boyu Zhu, Sicong Jiang, Zihan Wang, **Minglai Yang**, Zhe Zhao, Muhao Chen
    arXiv preprint, 2026 [preprint]
14. *Word2VecGD: Neural Graph Drawing with Cosine-Stress Optimization*
    Minglai Yang, Reyan Ahmed
    arXiv preprint, 2025 [preprint]

Honors and Awards

2025: **Best Senior Award**, *University of Arizona*

2025: **2nd Place — Reddit Wildcat Hackathon** *Reddit & University of Arizona*

2025: **Undergraduate Research Travel Grant (Top 10 awardees university-wide)**, *University of Arizona*

2025: **Galileo Circle Scholar (Top 0.8% Academic Award)**, College of Science, *University of Arizona*

2023–2025: **Academic Highest Distinction (Dean's List) (6 times)**, *University of Arizona*

2023–2025: **Global Wildcat Scholarship**, *University of Arizona*

2021: **Shanghai High School Science and Technology Competition**, Second Prize, *Shanghai High School*

2020: **Shanghai Youth Science and Technology Innovation Competition**, Third Prize, *Shanghai, China*

Leadership and Extracurricular Activities

AI Club, University of Arizona **Tucson, AZ**

President *Aug 2024 – Dec 2025*

- Advisor: Dr. Zhuolin Yang, Dr. Adarsh Pyarelal and Prof. Kobus Barnard
- Organized workshops on AI agents and LLM applications, including a Hack Arizona event with 100+ participants.
- Led weekly lectures and invited speaker sessions, engaging over 40 students across disciplines to foster AI learning.
- Organized a reading group that mentors students interested in AI research.
- Raised over $14,000 in sponsorships to support club activities and student research projects.

Mathematical Contest in Modeling (MCM), University of Arizona **Tucson, AZ**

Team Leader & First Author *Sep 2023 – Mar 2024*

- Led weekly tutoring sessions on mathematical modeling, convex optimization, Bayesian inference, and deep neural network methods.
- Developed mathematical models for the 2024 MCM problem on sea lamprey ecosystems, focusing on resource availability and sex ratio dynamics.

Teaching

Fall 2025: TA, CSC-244: Discrete Mathematics II, Computer Science, University of Arizona

Fall 2025: Instructor, Deep Learning and Natural Language Processing, AI Club, University of Arizona

Spring 2025: TA, CSC-144: Discrete Mathematics I, Computer Science, University of Arizona

Fall 2024 – Spring 2025: Instructor, Math for AI Workshop Series, AI Club, University of Arizona

Service

ACL Reviewer *2026*

ICML Reviewer *2026*

BlackboxNLP Workshop @ EMNLP Reviewer *2026*

AI4MATH Workshop @ ICML Reviewer *2024*

Technical Skills

Languages & Systems: Python, Java, C/C++, JavaScript, MATLAB, Ruby, Swift; Linux, Git, Docker, AWS, Nginx

ML/DL & Finetuning: PyTorch, NumPy, pandas, scikit-learn, PyMC; PEFT (LoRA/QLoRA, Prefix/P-Tuning), SFT/DPO/PPO, RLHF/RLAIF; LLaMA-Factory, VERL, OpenRLHF, DeepSpeed-Chat; bitsandbytes

NLP & RAG/Serving: Hugging Face (Transformers/Datasets/Tokenizers), spaCy, NLTK, SentencePiece, Sentence-Transformers; LangChain, LlamaIndex, FAISS, Milvus, Qdrant, Weaviate, Chroma, pgvector; Elasticsearch/BM25, vLLM, FlashAttention, TransformerLens